GoldDigger: Echoes of Conflict

A Data-Driven Analysis of Global Terrorism (1970–2017)

Source: Global Terrorism Database (START Center, University of Maryland)

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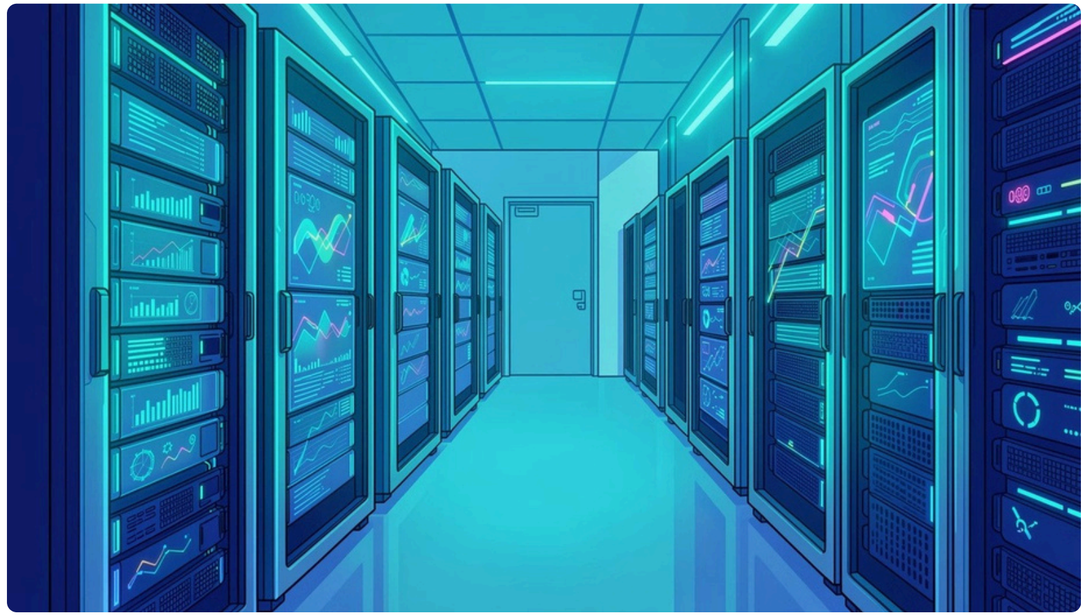
Project Overview & Dataset

The Global Terrorism Database

Maintained by the START Center at the University of Maryland, the GTD represents the largest open-source terrorism database in the world, providing comprehensive documentation of terrorist incidents across five decades.



Raw Dataset

181,691 incidents

135 columns 1970–2017 coverage

Cleaned Dataset

34,268 incidents

9 columns

High-confidence subset

Selected Features

Year, Month, Duration

Country, Region, City

Crime, Reason, Individual

The data pipeline followed a systematic approach: **Extraction → Cleaning → Exploratory Data Analysis → Statistical Analysis**. The cleaning process removed 147,423 rows (81%) due to missing motive or city data, resulting in a well-documented, high-confidence subset for analysis.

Data Cleaning & Preprocessing

From181,691raw incidents with135 columns,thepreprocessingpipelinesystematicallyselected 9 meaningful features and applied rigorous quality controls to ensure data integrity.

1

Initial State

181,691 rows × 135 columns

Raw GTD export

2

Column Selection

9 meaningful columns retained

126 dropped

3

Missing Value Processing

Empty strings → NaN

"Unknown" → NaN

4

Final Dataset

34,268 rows × 9 columns

Complete cases only

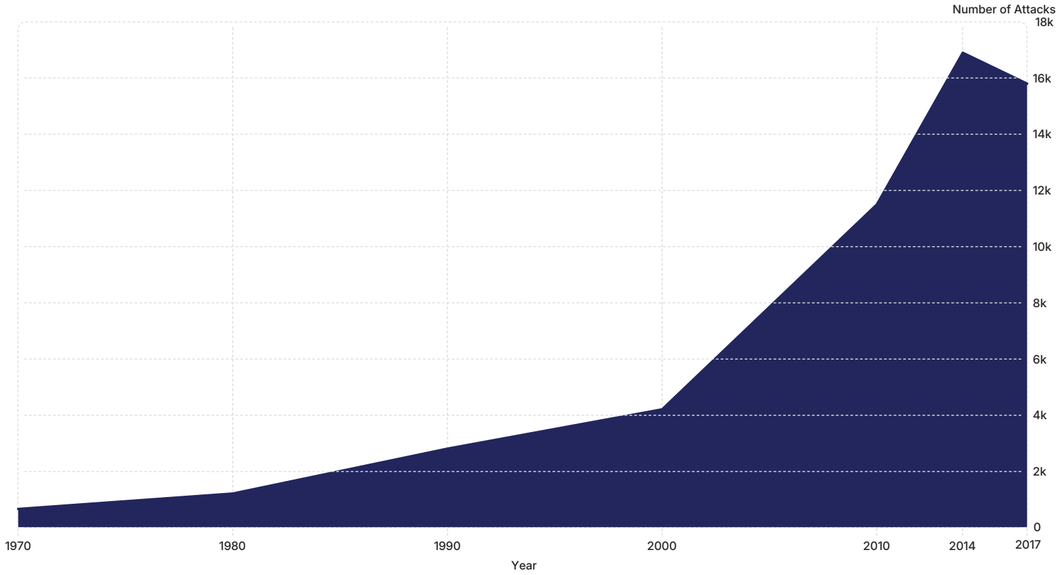
Missing Value Breakdown

- Reason/Motive: 72.17% missing (primary cause of row loss)
- City: 0.24% missing
- All other selected columns: 0% missing
- No duplicate event IDs found in raw data

⚠ The high missing rate in Reason (72%) introduces survivorship bias—findings apply primarily to incidents where motives were explicitly documented.

Temporal Trends: 1970–2017

Terroristattacksgrew from651incidentsperyearin1970toa peakof16,903in2014—an increase of 2,497% over 47 years. The Mann-Kendall Trend Test confirmed this upward trajectory is statistically significant.



16,903

Highest single-year attack count

0.53

Strong positive trend coefficient

<0.001

Statistically significant ($p < 0.001$)

+594%

Biggest YoY increase (1998: +558%)

Key Observations

- 1993 data missing from GTD
- Post-2014 slight decline but historically high levels
- Top decline years: 1979 (−69%), 1972 (−65%), 2012 (−57%)

Historical Context

The 2014 peak coincided with the rise of ISIS/ISIL and escalating conflicts in Iraq, Syria, and Afghanistan. The statistically confirmed trend ($\tau = 0.53$, $p < 0.001$) indicates terrorism is not randomly distributed over time but follows a consistent upward trajectory.

Geographic Hotspots

Attacksare heavilyconcentrated in specific regionsandcountries. The top two regions account for 65.9% of all incidents, while Iraq alone represents 18.4% of the cleaned dataset.

Top 5 Countries by Attack Count

1.**Iraq:** 6,320 attacks (Z-score = 8.1 — extreme outlier)

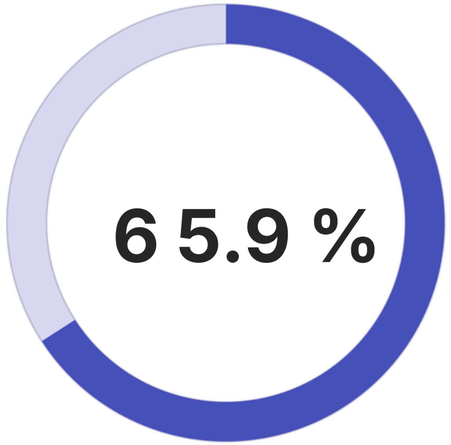
2. **India:** 4,363 attacks (Z = 5.5)
3. **Pakistan:** 4,227 attacks (Z = 5.3)

4.**Afghanistan:** 2,234 attacks (Z = 2.7)

5. **Philippines:** 1,839 attacks (Z = 2.2)

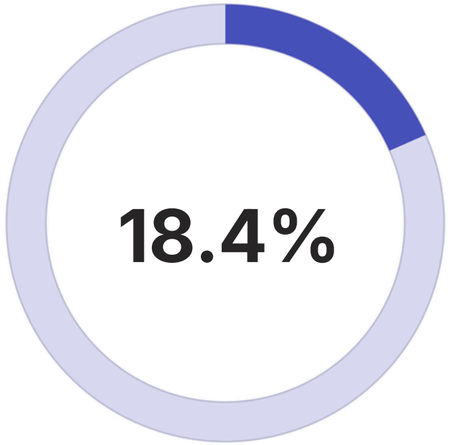
Top Regions by Share

- **South Asia:** 36.45% (12,491)
- **Middle East & North Africa:** 29.48% (10,103)
- **Southeast Asia:** 9.95%
- **Sub-Saharan Africa:** 8.95%
- **Top 2 regions = 65.9% of all attacks**



Top 2 Regions

Share of all attacks



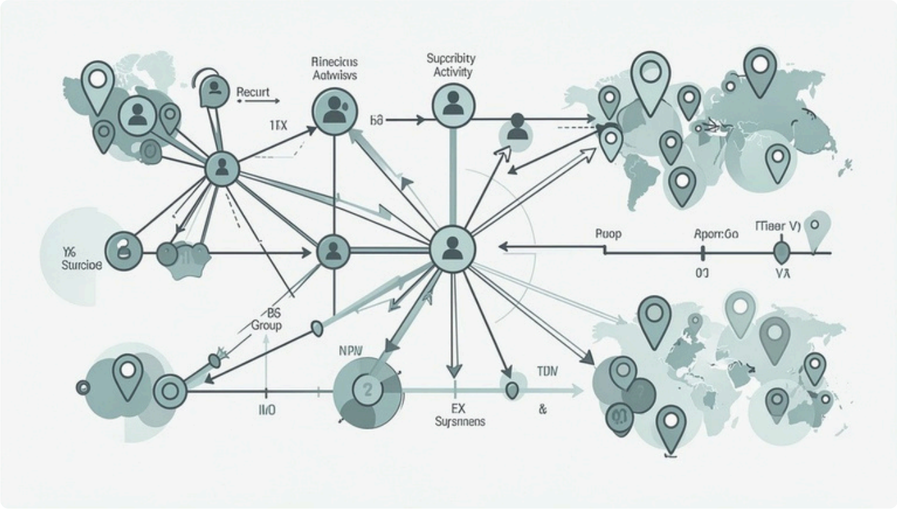
Iraq Alone

Of cleaned incidents

Iraq Z-score

Extreme outlier

Attack Characteristics & Patterns



Key Statistical Findings

The dataset reveals consistentpatterns in

how terrorist attacks are executed, who carries them out, and when they occur.

Terrorism Criteria

98.6% ofincidents(Crime= 1) meet formal terrorism classification criteria

Group Phenomenon

98.94%ofattackscarriedout by groups—only 1.06% by lone actors

Extended Duration

7.72%of attackslasted>24 hours; rate rose to 17.34% in 2016

Monthly Distribution Analysis

Attacks are spread fairly evenlyacrossall 12 months with no discernible seasonal pattern. The Kruskal-Wallis test on monthly attack counts yielded H = 3.88, p = 0.973, confirming **no significant seasonal effect**.

The absence of seasonal patterns suggests terrorism is driven by political and ideological factors rather than weather or calendar cycles.

Statistical Hypothesis Testing

Sixstatisticaltestswereconducted toidentifysignificantpatternsandrelationshipsinthe data. Five out of six tests returned statistically significant results ($p < 0.05$).

Test	Statistic	P-value	Signif ic ant?
Temporal Trend (Kendall)	$\tau = 0.5337$	<0.001	Yes
Region vs Duration (χ^2)	$\chi^2 = 701.20$	<0.001	Yes
Region vs Individual (χ^2)	$\chi^2 = 3763.8$	<0.001	Yes
Monthly Seasonality (K-W)	$H = 3.88$	0.973	No
Duration vs Individual (χ^2)	$\chi^2 = 21.69$	<0.001	Yes
Crime vs Individual (χ^2)	$\chi^2 = 37.73$	<0.001	Yes

Strong Regional Effects

Chi-square statistics in the thousands (Region vs Individual = 3,763.8) indicate extremely strong associations. Region strongly predicts both attack duration and lone-actor status.

No Seasonal Pattern

The only non-significant finding—monthly seasonality—is itself an important insight: attack timing is not calendar-driven but ideologically motivated.

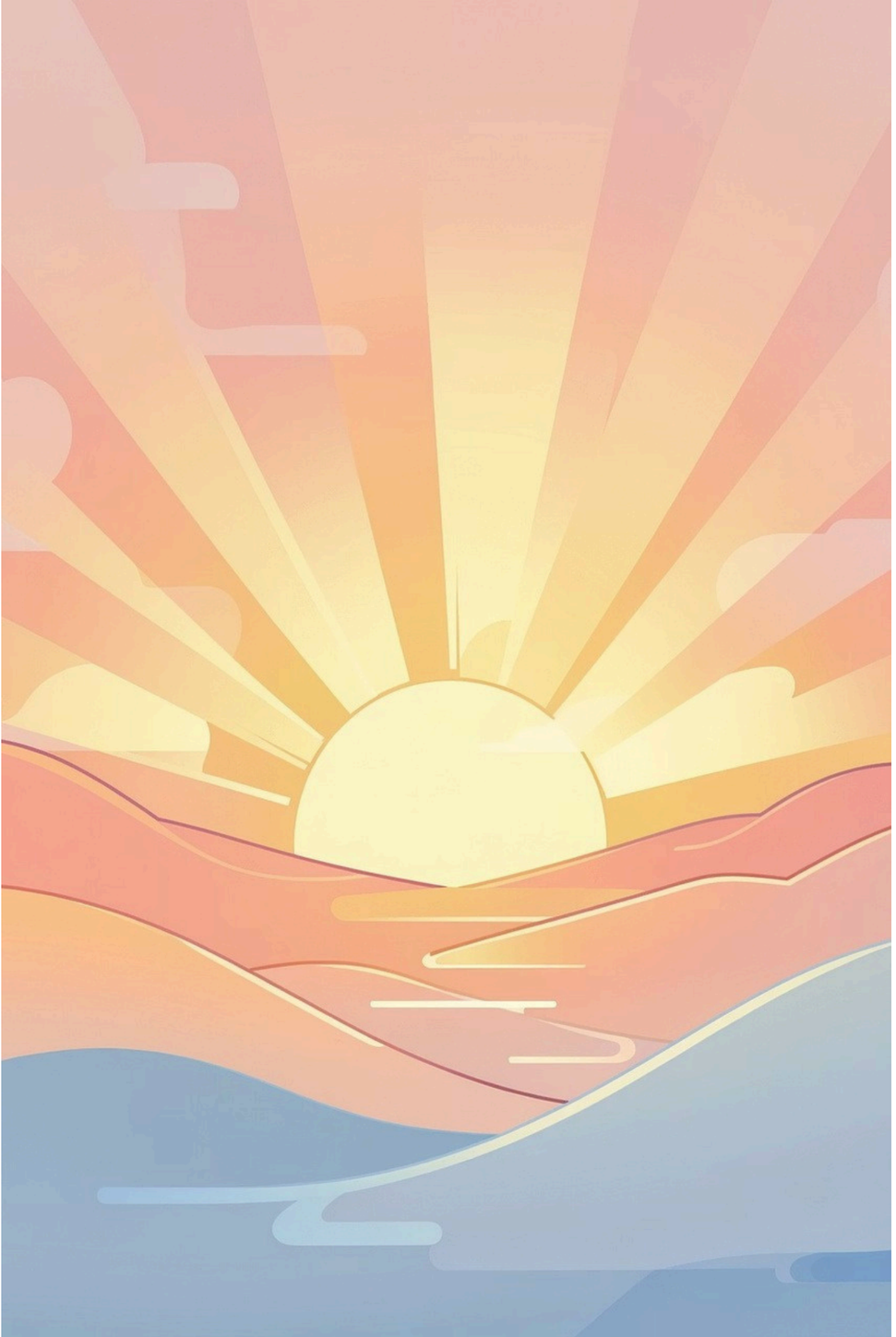
Limitations & Future Work

Current Limitations

- **Survivorship Bias:** 81% of raw rows dropped due to missing motive data (147,423 incidents lost)
- **Subset Analysis:** Findings based on 34,268 cleaned incidents may not generalize to full 181,691
- **No Casualty Data:** Fatalities (nkill) and injuries (nwound) excluded from cleaned pipeline

Future Research Directions

1	**NLP & Topic Modeling** Apply naturallanguageprocessing to Reason column text to cluster and categorize attack motives automatically
2	**Predictive Modeling** Build RandomForest,XGBoost, or neural network models to forecast attack likelihood by region and year
3	**Geospatial Mapping** Integrate latitude/longitudecoordinates from raw GTD for detailed geographic visualization and cluster analysis
4	**Post-2017 Updates** Incorporate latestGTDreleases to extend analysis through 2023 for current relevance
5	**Severity Analysis** Include casualtydata(fatalities, injuries) as additional outcome variable to assess attack impact



Thank You

KeyTakeaway: Behindeverydata point is a human story. Statistical analysis revealspatterns, butunderstanding the human impact remains paramount.

This analysis demonstrates that terrorism follows statistically identifiable patterns across time, geography, and attack type. The findings confirm a significant increase over 47 years, extreme geographic concentration, and the overwhelming role of group-based coordination.

Source: Global Terrorism Database, START Center, University of Maryland